

Lecture 10: steepest descent methods for multi-variable unconstrained optimization problem

MA3236 Non-Linear Programming

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Quick review

At $\mathbf{x}^{(k)}$, Newton-type methods start from

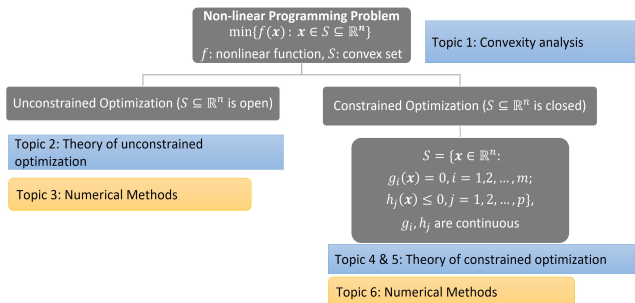
$$m^{(k)}(\mathbf{d}) = f(\mathbf{x}^{(k)}) + (\mathbf{g}^{(k)})^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T \mathbf{B}^{(k)} \mathbf{d}.$$

- **Pure Newton:** $\mathbf{B}^{(k)} = \mathbf{H}_f(\mathbf{x}^{(k)})$.
- **Shifted Newton:** $\mathbf{B}^{(k)} = \mathbf{H}_f(\mathbf{x}^{(k)}) + \tau^{(k)} \mathbf{I} \succ 0$.
- **Newton–Armijo:** safeguard until $(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)} < 0$, then backtrack; the safeguard itself does not require positive definiteness.

Typical line-search pattern

- Choose a descent direction $\mathbf{d}^{(k)}$ such that $(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)} < 0$.
- Select a step size $\alpha^{(k)}$ by exact line search or sufficient decrease.

Topics



In Topic 3, we will study numerical methods for solving **unconstrained optimization problem**.

- Methods for solving one-variable unconstrained optimization.
- Methods for solving multi-variable unconstrained optimization: Newton method, **steepest descent method**, conjugate-gradient method

Steepest descent direction: the identity model

Replace the Hessian by $\mathbf{I}$:

$$m^{(k)}(\mathbf{d}) = f(\mathbf{x}^{(k)}) + (\mathbf{g}^{(k)})^T \mathbf{d} + \frac{1}{2} \|\mathbf{d}\|^2.$$

Its Hessian with respect to $\mathbf{d}$ is $\mathbf{I} \succ 0$. Hence the model is strictly convex and has the unique minimizer

$$\nabla_{\mathbf{d}} m^{(k)}(\mathbf{d}) = \mathbf{g}^{(k)} + \mathbf{d} = \mathbf{0} \quad \implies \quad \mathbf{d}^{(k)} = -\mathbf{g}^{(k)}.$$

No Hessian evaluation or linear-system solve is needed.

Steepest descent direction: the fastest local speed

For a unit vector $\mathbf{u}$, the directional derivative at $\mathbf{x}$ is

$$D_{\mathbf{u}}f(\mathbf{x}) = \nabla f(\mathbf{x})^T \mathbf{u}.$$

It measures the first-order change per unit distance:

$$f(\mathbf{x} + t\mathbf{u}) = f(\mathbf{x}) + t\nabla f(\mathbf{x})^T \mathbf{u} + o(t).$$

- Negative directional derivative means local decrease.
- More negative means faster first-order decrease per unit distance.

Steepest descent direction: the fastest local speed (cont'd)

Among all Euclidean unit directions,

$$\min_{\|\mathbf{u}\|=1} \nabla f(\mathbf{x})^T \mathbf{u} = -\|\nabla f(\mathbf{x})\|.$$

Cauchy–Schwarz gives the bound, and equality holds at

$$\mathbf{u}_* = -\frac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|} \quad \text{when } \nabla f(\mathbf{x}) \neq \mathbf{0}.$$

Thus $-\nabla f(\mathbf{x})$ points in the same direction and is called the **steepest descent (SD) direction** in the Euclidean norm.

SD direction is a descent direction

If $\mathbf{g}^{(k)} \neq \mathbf{0}$ and $\mathbf{d}^{(k)} = -\mathbf{g}^{(k)}$, then

$$(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)} = -\|\mathbf{g}^{(k)}\|^2 < 0.$$

Differentiability gives

$$f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}) = f(\mathbf{x}^{(k)}) - \alpha \|\mathbf{g}^{(k)}\|^2 + o(\alpha).$$

Hence every sufficiently small $\alpha > 0$ decreases f .

SD with line search

Choose $\mathbf{x}^{(0)} \in \mathcal{D}$, $\varepsilon_g > 0$, and a maximum number K of updates. For $k = 0, \dots, K - 1$:

- 1 Compute $\mathbf{g}^{(k)} = \nabla f(\mathbf{x}^{(k)})$. Stop if $\|\mathbf{g}^{(k)}\| \leq \varepsilon_g$.
- 2 Set $\mathbf{d}^{(k)} = -\mathbf{g}^{(k)}$.
- 3 Choose $\alpha^{(k)} > 0$ by exact line search or Armijo backtracking.
- 4 Update

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)}.$$

Check the stopping test after the last permitted update.

Python code: `steepest_descent(...)` in `L10_steepest_descent.py`

Exact and Armijo line search

For $\phi^{(k)}(\alpha) = f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)})$:

Exact line search

$$\alpha^{(k)} \in \arg \min_{\alpha \geq 0} \phi^{(k)}(\alpha).$$

It chooses the best point on the current ray, when the minimum exists.

Armijo backtracking

Choose $\bar{\alpha} > 0$ and $\beta, \sigma \in (0, 1)$. Test $\alpha = \bar{\alpha}, \beta\bar{\alpha}, \beta^2\bar{\alpha}, \dots$ until

$$f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}) \leq f(\mathbf{x}^{(k)}) + \sigma \alpha (\mathbf{g}^{(k)})^T \mathbf{d}^{(k)}.$$

It asks for sufficient decrease rather than the line minimum.

Exact line search on a strictly convex quadratic

For $\phi^{(k)}(\alpha) = f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)})$:

Exact line search

$$\alpha^{(k)} \in \arg \min_{\alpha \geq 0} \phi^{(k)}(\alpha).$$

If

$$f(\mathbf{x}) = q(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} - \mathbf{c}^T \mathbf{x}, \quad \mathbf{Q} \succ 0, \quad \mathbf{g}^{(k)} = \mathbf{Q} \mathbf{x}^{(k)} - \mathbf{c}.$$

Along $-\mathbf{g}^{(k)}$,

$$\phi^{(k)}(\alpha) = q(\mathbf{x}^{(k)}) - \alpha \|\mathbf{g}^{(k)}\|^2 + \frac{\alpha^2}{2} (\mathbf{g}^{(k)})^T \mathbf{Q} \mathbf{g}^{(k)}.$$

Exact line search on a strictly convex quadratic

For $\phi^{(k)}(\alpha) = f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)})$:

Exact line search

$$\alpha^{(k)} \in \arg \min_{\alpha \geq 0} \phi^{(k)}(\alpha).$$

If

$$f(\mathbf{x}) = q(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} - \mathbf{c}^T \mathbf{x}, \quad \mathbf{Q} \succ 0, \quad \mathbf{g}^{(k)} = \mathbf{Q} \mathbf{x}^{(k)} - \mathbf{c}.$$

Along $-\mathbf{g}^{(k)}$,

$$\phi^{(k)}(\alpha) = q(\mathbf{x}^{(k)}) - \alpha \|\mathbf{g}^{(k)}\|^2 + \frac{\alpha^2}{2} (\mathbf{g}^{(k)})^T \mathbf{Q} \mathbf{g}^{(k)}.$$

Therefore

$$\alpha^{(k)} = \frac{\|\mathbf{g}^{(k)}\|^2}{(\mathbf{g}^{(k)})^T \mathbf{Q} \mathbf{g}^{(k)}}.$$

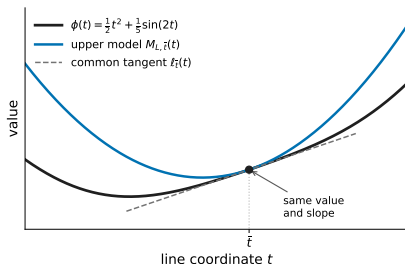
Armijo backtracking on an L -smooth function

Let $\mathcal{D} \subseteq \mathbf{R}^n$ be open and convex. A differentiable f is L -**smooth** on $\mathcal{D}$ if

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\|, \quad \mathbf{x}, \mathbf{y} \in \mathcal{D}.$$

This implies, by the fundamental theorem of calculus, if $\mathbf{x}, \mathbf{y} \in \mathcal{D}$,

$$f(\mathbf{x} + \mathbf{d}) \leq M_{L,\mathbf{x}}(\mathbf{d}) := f(\mathbf{x}) + \nabla f(\mathbf{x})^T \mathbf{d} + \frac{L}{2}\|\mathbf{d}\|^2, \quad \mathbf{d} := \mathbf{y} - \mathbf{x}.$$



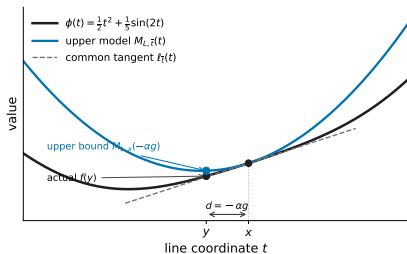
Armijo backtracking on an L -smooth function (cont'd)

Let

$$\mathbf{d} := -\alpha \nabla f(\mathbf{x}), \quad \mathbf{y} := \mathbf{x} + \mathbf{d} = \mathbf{x} - \alpha \nabla f(\mathbf{x}).$$

The same graph shows that the actual value lies below the upper model:

$$\begin{aligned} f(\mathbf{x} - \alpha \nabla f(\mathbf{x})) &\leq M_{L,\mathbf{x}}(-\alpha \nabla f(\mathbf{x})) = f(\mathbf{x}) - \alpha \|\nabla f(\mathbf{x})\|^2 + \frac{L\alpha^2}{2} \|\nabla f(\mathbf{x})\|^2 \\ &= f(\mathbf{x}) - \alpha \left(1 - \frac{L\alpha}{2}\right) \|\nabla f(\mathbf{x})\|^2. \end{aligned}$$



Armijo backtracking on an L -smooth function (cont'd)

$$f(\mathbf{x} - \alpha \nabla f(\mathbf{x})) \leq f(\mathbf{x}) - \alpha \left(1 - \frac{L\alpha}{2}\right) \|\nabla f(\mathbf{x})\|^2.$$

The right-hand side is minimized when $\alpha = 1/L$, for which the Armijo condition holds with $\sigma = 1/2$:

$$f\left(\mathbf{x} - \frac{1}{L} \nabla f(\mathbf{x})\right) \leq f(\mathbf{x}) - \frac{1}{2L} \|\nabla f(\mathbf{x})\|^2. \quad (\text{Descent inequality})$$

Armijo condition

$$f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}) \leq f(\mathbf{x}^{(k)}) + \sigma \alpha (\mathbf{g}^{(k)})^T \mathbf{d}^{(k)}.$$

A quadratic family with controlled shape

Let

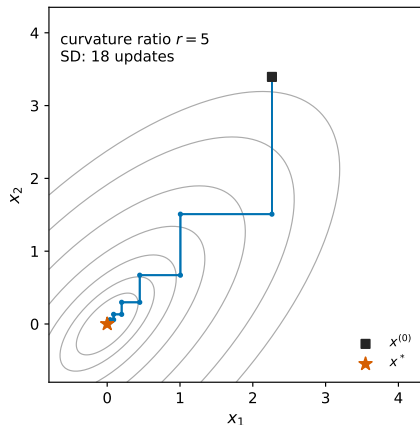
$$q_r(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q}_r \mathbf{x}, \quad \mathbf{Q}_r = \mathbf{U} \begin{pmatrix} 1 & 0 \\ 0 & r \end{pmatrix} \mathbf{U}^T, \quad r \geq 1,$$

where $\mathbf{U} \in \mathbf{R}^{2 \times 2}$ is a fixed rotation.

- The unique minimizer is $\mathbf{x}^* = \mathbf{0}$.
- The eigenvalues are 1 and r .
- Larger r makes the contours more elongated.

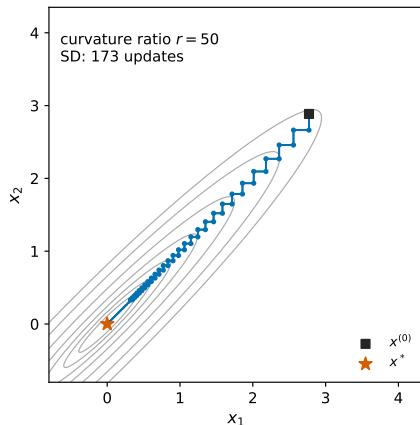
We use the same eigen-coordinate initialization $\mathbf{U}^T \mathbf{x}^{(0)} = (4, 4/r)^T$.

SD with exact line search when $r = 5$



The exact line search crosses the valley at right angles. It takes 18 updates to reach relative objective gap at most 10^{-6} .

SD with exact line search when $r = 50$



It now takes 173 updates to reach relative objective gap at most 10^{-6} .
Exact line search optimizes each line, but it does not remove zig-zagging.

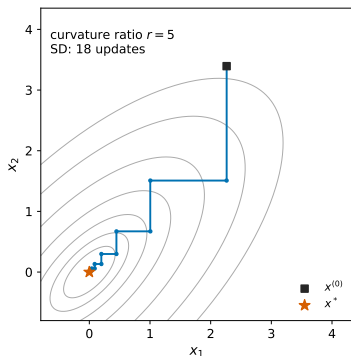
Orthogonality under exact line search

Let $\alpha^{(k)}$ be an attained exact line minimizer,

$$\alpha^{(k)} \in \arg \min_{\alpha \geq 0} \phi^{(k)}(\alpha) = f(\mathbf{x}^{(k)} - \alpha \mathbf{g}^{(k)}),$$

and suppose $\mathbf{g}^{(k)} \neq \mathbf{0}$. Then

$$(\mathbf{g}^{(k+1)})^T \mathbf{g}^{(k)} = 0.$$

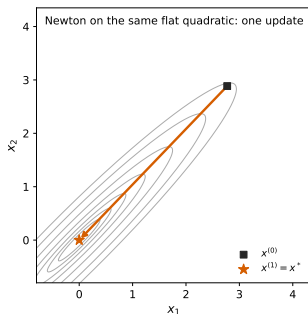


Newton solves every strictly convex quadratic in one step

For $q(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x} - \mathbf{c}^T\mathbf{x}$,

$$\mathbf{d}_N = -\mathbf{Q}^{-1}(\mathbf{Q}\mathbf{x} - \mathbf{c}) = \mathbf{x}^* - \mathbf{x}.$$

Thus the full Newton update is $\mathbf{x}^+ = \mathbf{x}^*$, from every starting point.

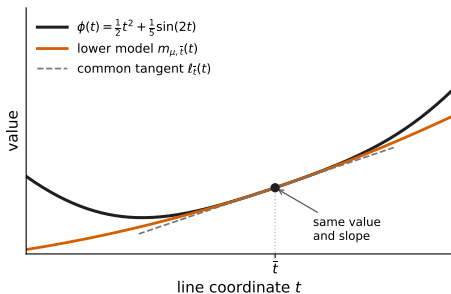


The Hessian records both the steep and flat directions. The identity model treats them as if their curvature were equal.

Strong convexity

A differentiable $f: \mathcal{D} \rightarrow \mathbf{R}$, with $\mathcal{D}$ convex, is μ -**strongly convex** if $\mu > 0$ and

$$f(\mathbf{x} + \mathbf{d}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T \mathbf{d} + \frac{\mu}{2} \|\mathbf{d}\|^2 =: m_{\mu, \mathbf{x}}(\mathbf{d}) \quad \forall \mathbf{x}, \mathbf{y} \in \mathcal{D}, \mathbf{d} := \mathbf{y} - \mathbf{x}.$$



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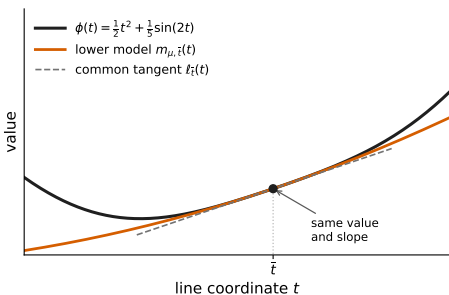
$$f(\mathbf{x} + \mathbf{d}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T \mathbf{d} + \frac{\mu}{2} \|\mathbf{d}\|^2 =: m_{\mu, \mathbf{x}}(\mathbf{d}) \quad \forall \mathbf{x}, \mathbf{y} \in \mathcal{D}, \mathbf{d} := \mathbf{y} - \mathbf{x}.$$

strongly convex $\implies$ strictly convex $\implies$ convex.

The converses fail on $\mathbf{R}$:

- $f(t) = t^2$: strongly convex;
- $f(t) = t^4$: strictly, but not strongly, convex;
- $f(t) = 0$: convex, but not strictly convex.

Only strong convexity supplies a uniform quadratic gap $\mu > 0$.



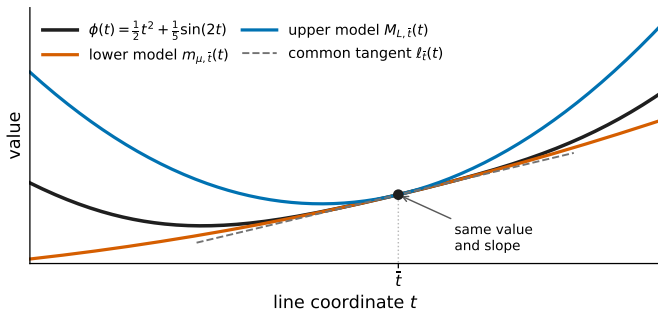
Lower and upper curvature bounds

At the same base point $\mathbf{x}$, μ -strong convexity and L -smoothness give

$$m_{\mu,\mathbf{x}}(\mathbf{d}) \leq f(\mathbf{x} + \mathbf{d}) \leq M_{L,\mathbf{x}}(\mathbf{d}).$$

If $f \in \mathcal{C}^2(\mathcal{D})$, the corresponding curvature bounds are

$$\mu \mathbf{I} \preceq \mathbf{H}_f(\mathbf{x}) \preceq L \mathbf{I} \quad \forall \mathbf{x} \in \mathcal{D}.$$



SD: linear convergence with μ, L

Let $f: \mathbf{R}^n \rightarrow \mathbf{R}$ be differentiable, μ -strongly convex, and L -smooth, and let $\mathbf{x}^*$ be its unique minimizer. For either $\alpha^{(k)} = 1/L$ or attained exact line search,

$$f(\mathbf{x}^{(k+1)}) \leq f(\mathbf{x}^{(k)}) - \frac{1}{2L} \|\mathbf{g}^{(k)}\|^2.$$

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Define the objective gap $E(\mathbf{x}) := f(\mathbf{x}) - f(\mathbf{x}^*)$ and $E^{(k)} := E(\mathbf{x}^{(k)})$. Then

$$E^{(k+1)} \leq E^{(k)} - \frac{1}{2L} \|\mathbf{g}^{(k)}\|^2.$$

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$$E^{(k+1)} \leq E^{(k)} - \frac{1}{2L} \|\mathbf{g}^{(k)}\|^2.$$

By μ -strong convexity,

$$f(\mathbf{x}^*) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{x}^* - \mathbf{x}) + \frac{\mu}{2} \|\mathbf{x}^* - \mathbf{x}\|^2 \Rightarrow \|\nabla f(\mathbf{x})\|^2 \geq 2\mu E(\mathbf{x}).$$

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Hence

$$E^{(k+1)} \leq \left(1 - \frac{\mu}{L}\right) E^{(k)} \Rightarrow E^{(k)} \leq \left(1 - \frac{\mu}{L}\right)^k E^{(0)}.$$

SD: linear convergence with μ, L (cont'd)

Let $\rho = 1 - \mu/L$. Strong convexity and smoothness imply

$$\|\mathbf{x}^{(k)} - \mathbf{x}^*\| \leq \sqrt{\frac{2E^{(0)}}{\mu}} \rho^{k/2},$$

and

$$\|\mathbf{g}^{(k)}\| \leq \sqrt{2LE^{(0)}} \rho^{k/2}.$$

Thus both quantities have geometric upper bounds of order $\rho^{k/2}$.

Condition number and iteration complexity

Using the tight constants, define the condition number of f

$$\kappa_f = \frac{L}{\mu} \geq 1.$$

Then the function-gap factor is

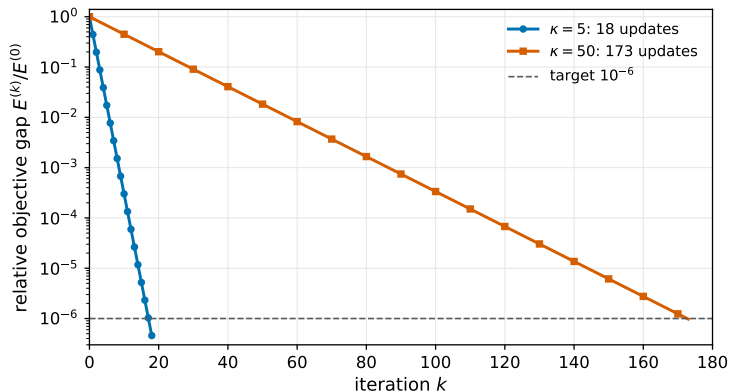
$$\rho = 1 - \frac{1}{\kappa_f}.$$

For $\kappa_f > 1$, to guarantee $E^{(k)} \leq \varepsilon E^{(0)}$, it is sufficient that

$$k \geq \frac{\log(\varepsilon^{-1})}{\log((1 - 1/\kappa_f)^{-1})} = O(\kappa_f \log(\varepsilon^{-1})).$$

A large condition number predicts slow worst-case progress.

Condition number and observed convergence



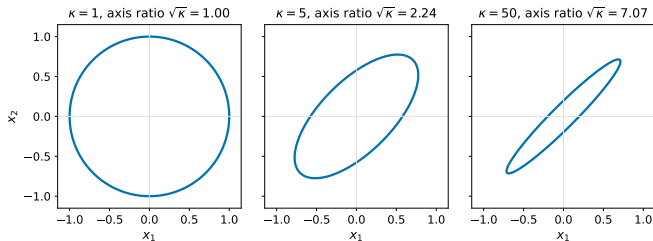
For the controlled quadratic family, the flatter landscape gives a gap ratio closer to one. Newton still reaches the minimizer in one update.

Condition number is visible in the landscape

For $q(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x}$, $\mathbf{Q} \succ 0$,

$$L = \lambda_{\max}(\mathbf{Q}), \quad \mu = \lambda_{\min}(\mathbf{Q}), \quad \kappa_f = \frac{\lambda_{\max}(\mathbf{Q})}{\lambda_{\min}(\mathbf{Q})}.$$

When $n = 2$, the ratio of the long to short semiaxis of a level curve is $\sqrt{\kappa_f}$.



Preconditioned SD

For $q(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x}$ with $\mathbf{Q} \succ 0$, choose a **fixed** $\mathbf{B} \succ 0$ and set

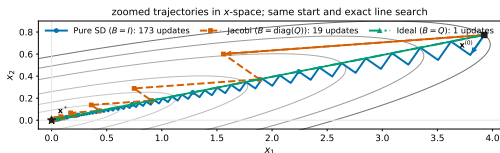
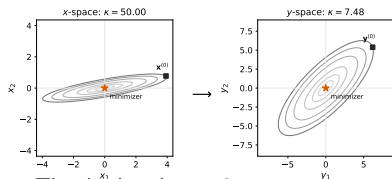
$$\mathbf{y} := \mathbf{B}^{1/2}\mathbf{x}, \quad \tilde{\mathbf{Q}} := \mathbf{B}^{-1/2}\mathbf{Q}\mathbf{B}^{-1/2}.$$

Then $q(\mathbf{x}) = \tilde{q}(\mathbf{y}) = \frac{1}{2}\mathbf{y}^T\tilde{\mathbf{Q}}\mathbf{y}$. Ordinary SD in $\mathbf{y}$ induces the following preconditioned update in $\mathbf{x}$:

$$\mathbf{y}^{(k+1)} = \mathbf{y}^{(k)} - \alpha^{(k)}\tilde{\mathbf{Q}}\mathbf{y}^{(k)}, \quad \mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \alpha^{(k)}\mathbf{B}^{-1}\mathbf{Q}\mathbf{x}^{(k)}.$$

The relevant condition number is

$$\kappa_{\text{eff}}(\mathbf{Q}; \mathbf{B}) := \kappa(\tilde{\mathbf{Q}}) = \frac{\lambda_{\max}(\tilde{\mathbf{Q}})}{\lambda_{\min}(\tilde{\mathbf{Q}})}.$$



The displayed $r = 50$ experiment uses a separate 10° -rotated quadratic and $\mathbf{B}_J = \text{diag}(\mathbf{Q})$.

Preconditioned SD: the modified local model

For a general smooth f , $-\mathbf{B}^{-1}\mathbf{g}^{(k)}$ is the unique minimizer of

$$m_{\mathbf{B}}^{(k)}(\mathbf{d}) := f(\mathbf{x}^{(k)}) + (\mathbf{g}^{(k)})^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T \mathbf{B} \mathbf{d}, \quad \mathbf{B} \succ 0.$$

Its stationarity equation is

$$\mathbf{B} \mathbf{d}^{(k)} = -\mathbf{g}^{(k)}, \quad \mathbf{d}^{(k)} = -\mathbf{B}^{-1} \mathbf{g}^{(k)}.$$

If $\mathbf{g}^{(k)} \neq \mathbf{0}$, then

$$(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)} = -(\mathbf{g}^{(k)})^T \mathbf{B}^{-1} \mathbf{g}^{(k)} < 0.$$

Thus $\mathbf{d}^{(k)}$ is a descent direction, and a line search chooses its length:

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)}.$$

What if strong convexity is missing?

Consider the one-dimensional function

$$\varphi(x) = e^{-x^2} + x^2 - 1.$$

It is convex and L -smooth, with

$$\varphi''(x) = 2 + (4x^2 - 2)e^{-x^2}, \quad L = 2 + 4e^{-3/2}.$$

But $\varphi''(0) = 0$, so no positive global strong-convexity constant exists. The unique minimizer is $x^* = 0$.

Question

Does smoothness alone still guarantee a Q-linear objective gap?

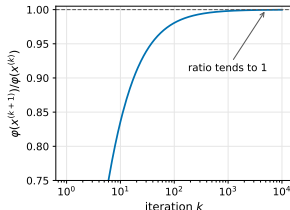
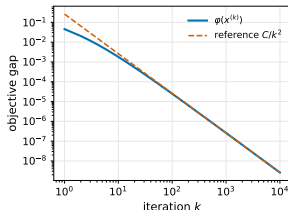
A smooth convex example with Q-sublinear decay

Use $\alpha = 1/L$ and $x^{(0)} = 1$. Near zero,

$$x^{(k+1)} = x^{(k)} - \frac{2}{L}(x^{(k)})^3 + O((x^{(k)})^5), \quad \varphi(x) = \frac{x^4}{2} + O(x^6).$$

Consequently,

$$\frac{\varphi(x^{(k+1)})}{\varphi(x^{(k)})} \rightarrow 1, \quad \varphi(x^{(k)}) = \Theta(k^{-2}).$$



What the counterexample shows

The descent inequality still guarantees decrease:

$$\varphi(x^{(k+1)}) \leq \varphi(x^{(k)}) - \frac{1}{2L} |\varphi'(x^{(k)})|^2.$$

The missing step is a uniform error bound

$$|\varphi'(x)|^2 \geq 2\mu(\varphi(x) - \varphi(0)) \quad \text{with } \mu > 0.$$

Near the minimizer, the landscape becomes too flat for such a constant.

Warning

Convexity + smoothness $\nRightarrow$ Q-linear convergence.

From SD to stochastic gradients

Machine-learning objectives often have the finite-sum form

$$F(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N \ell_i(\mathbf{w}), \quad \nabla F(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N \nabla \ell_i(\mathbf{w}).$$

The full gradient may be expensive when N is large.

Mini-batch idea

Sample a small batch $\mathcal{B}^{(k)}$ and replace the full gradient by

$$\hat{\mathbf{g}}^{(k)} = \frac{1}{|\mathcal{B}^{(k)}|} \sum_{i \in \mathcal{B}^{(k)}} \nabla \ell_i(\mathbf{w}^{(k)}).$$

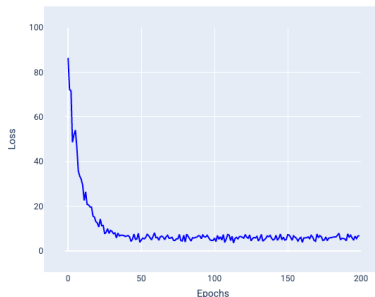
Under uniform sampling, this is an unbiased gradient estimate.

Stochastic gradient descent in practice

The stochastic-gradient update is

$$\mathbf{w}^{(k+1)} = \mathbf{w}^{(k)} - \eta^{(k)} \hat{\mathbf{g}}^{(k)}.$$

- **Smaller batches:** cheaper but noisier updates.
- **Learning-rate decay:** smaller late-stage fluctuations.
- **Momentum:** averages recent directions.
- **AdaGrad, RMSProp, Adam:** adaptive coordinatewise scaling.



Random sampling produces persistent loss fluctuations.

Source: Google Machine Learning Crash Course, Fig. 24 (CC BY 4.0).

Take-home messages

- SD from replacing Hessian by $\mathbf{I}$ or local fastest decrease.
- Exact or Armijo line search gives step sizes.
- Strong convexity and smoothness $\Rightarrow$ condition-sensitive rate.
- Preconditioning changes the metric and may improve the condition.
- Stochastic gradient descent is a the workhorse in deep learning.

Reference proof: the steepest unit direction

For every $\|\mathbf{u}\| = 1$, Cauchy–Schwarz gives

$$\nabla f(\mathbf{x})^T \mathbf{u} \geq -\|\nabla f(\mathbf{x})\| \|\mathbf{u}\| = -\|\nabla f(\mathbf{x})\|.$$

If $\nabla f(\mathbf{x}) \neq \mathbf{0}$, equality holds at

$$\mathbf{u}_* = -\frac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|}.$$

Moreover, every positive multiple $-c\nabla f(\mathbf{x})$ is a descent direction:

$$\nabla f(\mathbf{x})^T [-c\nabla f(\mathbf{x})] = -c\|\nabla f(\mathbf{x})\|^2 < 0.$$

Reference proof: exact-search orthogonality

Let $\phi(\alpha) = f(\mathbf{x}^{(k)} - \alpha \mathbf{g}^{(k)})$. Since

$$\phi'(0) = -\|\mathbf{g}^{(k)}\|^2 < 0,$$

an attained exact minimizer cannot be 0. It is therefore an interior minimizer and

$$0 = \phi'(\alpha^{(k)}) = -(\mathbf{g}^{(k+1)})^T \mathbf{g}^{(k)}.$$

Finally,

$$\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)} = -\alpha^{(k)} \mathbf{g}^{(k)}, \quad \mathbf{x}^{(k+2)} - \mathbf{x}^{(k+1)} = -\alpha^{(k+1)} \mathbf{g}^{(k+1)}.$$

Whenever both displacements are nonzero, they are orthogonal.

Reference proof: descent lemma

Set $\mathbf{d} = \mathbf{y} - \mathbf{x}$ and $h(t) = f(\mathbf{x} + t\mathbf{d})$. Then

$$\begin{aligned} f(\mathbf{y}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^T \mathbf{d} &= \int_0^1 [\nabla f(\mathbf{x} + t\mathbf{d}) - \nabla f(\mathbf{x})]^T \mathbf{d} \, dt \\ &\leq \int_0^1 Lt \|\mathbf{d}\|^2 \, dt \\ &= \frac{L}{2} \|\mathbf{d}\|^2. \end{aligned}$$

The first equality is the fundamental theorem of calculus. The inequality uses Cauchy–Schwarz and the L -Lipschitz continuity of the gradient.

Reference proof: the gradient-gap bounds

Strong convexity gives, for every $\mathbf{y}$,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) + \frac{\mu}{2}\|\mathbf{y} - \mathbf{x}\|^2.$$

Minimizing the right side over $\mathbf{y}$ gives

$$f(\mathbf{x}^*) \geq f(\mathbf{x}) - \frac{1}{2\mu}\|\nabla f(\mathbf{x})\|^2.$$

The descent lemma at $\mathbf{y} = \mathbf{x} - \nabla f(\mathbf{x})/L$, together with $f(\mathbf{x}^*) \leq f(\mathbf{y})$, gives

$$f(\mathbf{x}^*) \leq f(\mathbf{x}) - \frac{1}{2L}\|\nabla f(\mathbf{x})\|^2.$$

Rearranging yields $2\mu E(\mathbf{x}) \leq \|\nabla f(\mathbf{x})\|^2 \leq 2LE(\mathbf{x})$.

Reference proof: the general linear rate

The descent lemma at the trial step $1/L$ gives

$$E^{(k+1)} \leq E^{(k)} - \frac{1}{2L} \|\mathbf{g}^{(k)}\|^2$$

for the fixed step. Exact line search does at least as well as this trial. The gradient-gap bound gives

$$\|\mathbf{g}^{(k)}\|^2 \geq 2\mu E^{(k)}.$$

Hence

$$E^{(k+1)} \leq \left(1 - \frac{\mu}{L}\right) E^{(k)}.$$

Induction yields $E^{(k)} \leq (1 - \mu/L)^k E^{(0)}$.

Reference: strong-convexity characterizations

For differentiable f , the following statements are equivalent:

f is μ -strongly convex,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|^2,$$

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}) - \frac{\mu}{2} \lambda (1 - \lambda) \|\mathbf{x} - \mathbf{y}\|^2.$$

Apply the usual first-order and chord characterizations of convexity to

$$h(\mathbf{x}) = f(\mathbf{x}) - \frac{\mu}{2} \|\mathbf{x}\|^2.$$

Expanding h gives the two displayed inequalities.

Reference: exact-search quadratic error identity

For $q(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x} - \mathbf{c}^T\mathbf{x}$, let $E^{(k)} = q(\mathbf{x}^{(k)}) - q(\mathbf{x}^*)$ and assume $\mathbf{g}^{(k)} \neq \mathbf{0}$. Using the exact step and $E^{(k)} = \frac{1}{2}(\mathbf{g}^{(k)})^T\mathbf{Q}^{-1}\mathbf{g}^{(k)}$,

$$\frac{E^{(k+1)}}{E^{(k)}} = 1 - \frac{((\mathbf{g}^{(k)})^T\mathbf{g}^{(k)})^2}{((\mathbf{g}^{(k)})^T\mathbf{Q}\mathbf{g}^{(k)})(\mathbf{g}^{(k)})^T\mathbf{Q}^{-1}\mathbf{g}^{(k)}}.$$

This identity is sharper than the general smooth strongly convex estimate.

Reference: exact-search quadratic rate

The Kantorovich inequality gives

$$\frac{(\mathbf{z}^T \mathbf{z})^2}{(\mathbf{z}^T \mathbf{Q} \mathbf{z})(\mathbf{z}^T \mathbf{Q}^{-1} \mathbf{z})} \geq \frac{4\lambda_{\min} \lambda_{\max}}{(\lambda_{\min} + \lambda_{\max})^2}.$$

Applying it to $\mathbf{z} = \mathbf{g}^{(k)}$ yields

$$\frac{E^{(k+1)}}{E^{(k)}} \leq \left(\frac{\kappa_f - 1}{\kappa_f + 1} \right)^2.$$

This explicit factor is special to positive-definite quadratics with exact line search. The lecture notes contain the complete proof of the inequality.